

Boost episode-anchored impact estimate

What V5's dosing changes would have done at real LOW/HIGH episodes. Window: last **90 days**. Low <70.0, High >180.0 mg/dL. Users anonymised. Generated by `episode_impact.py`.

Method (and its limits — read first)

The real BG was driven by **V1**. For each episode we sum the dose V5 would have withheld vs V1 over the prior 150 min, weight each by how much of it would already have *acted* by the episode extreme (oref exponential insulin curve), and multiply by the user's logged **ISF**:

$$\Delta BG \approx ISF \times \sum (dose_V1 - dose_V5) \times activatedFraction(t_extreme - t_dose)$$

- **Low episodes:** +ΔBG = how much **shallower** the low would be under V5.
- **High episodes:** +ΔBG = how much **higher** the peak would run under V5 (the cost).
- **First-order, open-loop estimate — NOT a glucose simulation.** It adds only the marginal insulin effect and assumes carbs/counter-regulation/subsequent loop reactions are unchanged. Only shadow cycles (V1 drove the real BG) are attributed. Treat as order-of-magnitude, grounded in each user's real ISF.

Estimated impact at LOW episodes (V5 = gentler dosing → shallower lows)

user	days	ISF	lows	median nadir	median ΔBG	p75 ΔBG	lows lifted ≥ threshold
self	40	125	99	58 (3.2)	+32 (+1.8)	+90 (+5.0)	52/99 (53%)
A	24	36	16	61 (3.4)	+11 (+0.6)	+39 (+2.2)	7/16 (44%)
B	23	33	33	59 (3.3)	+22 (+1.2)	+90 (+5.0)	17/33 (52%)
C	13	51	25	61 (3.4)	+0 (+0.0)	+43 (+2.4)	9/25 (36%)
D	11	42	49	60 (3.3)	+19 (+1.1)	+39 (+2.2)	30/49 (61%)

ΔBG in mg/dL (mmol/L). “lifted” = nadir + estimated ΔBG would reach/exceed the low threshold — i.e. the episode would likely not have been a low under V5.

Estimated cost at HIGH episodes (V5 dosing less → higher peaks)

user	highs	median peak	median ΔBG (higher)	p75 ΔBG
self	103	207 (11.5)	+26 (+1.4)	+90 (+5.0)
A	60	200 (11.1)	+9 (+0.5)	+32 (+1.8)
B	82	203 (11.3)	+10 (+0.6)	+38 (+2.1)
C	14	199 (11.1)	+7 (+0.4)	+27 (+1.5)
D	3	187 (10.4)	+39 (+2.2)	+59 (+3.3)

Reading it

The trade V5 makes is visible directly: at lows it would have withheld insulin → a positive ΔBG (shallower / averted lows); at highs the same gentleness costs a positive ΔBG (higher peaks). If low-side ΔBG is materially larger than high-side ΔBG, V5 is trading a lot of low-risk for a little high — the intended behaviour. These are estimates; live multi-user shadow data is how we confirm them.